When ReAct Phase Probes Look Abstract: Cue-Dependent Decodability in Gemma-2 2B and 9B

Anonymous ACL submission

Abstract

Linear probes can decode the current phase of a ReAct trace with near-perfect accuracy, inviting an interpretation in terms of an abstract reasoning state. We test that interpretation in Gemma-2 base and instruction-tuned (IT) models at 2B and 9B. Natural-trace probes peak at layers 12–13 and remain near 0.98 accurate after markers and newlines are removed. A preregistered content-and-position-decorrelated dataset changes the conclusion. At 2B, marked traces retain substantial signal ($R_{\text{base}} = 0.630$, $R_{\text{IT}} = 0.778$; gap 0.148 $[0.123, 0.173]$), but removing markers reduces both models to $R \approx 0.28$ and shrinks the gap to 0.018 $[-0.006, 0.043]$. At 9B, IT–base invariance after cue removal replicates, with smaller point gaps but not tighter intervals: signed gaps are $+0.003$ and -0.006, with both intervals crossing zero. However, 9B straddles the strict $R \leq 0.33$ cutoff, and the large marked IT advantage does not replicate. Feature evidence at 2B corroborates content entanglement: 28 of 30 top phase-selective sparse-autoencoder slots are content-domain detectors. Bounded patching produces a small, L20-driven effect above a random-feature null. Thus high natural decodability is cue-dependent; the robust cross-scale result is IT/base invariance after cue removal, not a scale-universal abstract phase variable.

1 Introduction

Linear probing asks whether a property can be predicted from a model's hidden state. The method is useful precisely because it is permissive: a simple readout can reveal information distributed across many dimensions. That permissiveness also creates an interpretive hazard. A probe that decodes a label need not identify the variable named by the researcher; it may exploit any feature correlated with that label, and decodability alone does not establish that the model uses the decoded information (Hewitt and Liang, 2019; Ravichander et al., 2021; Elazar et al., 2021; Belinkov, 2022).

Reasoning traces make this hazard unusually sharp. ReAct alternates between labeled Thought, Action, and Observation segments (Yao et al., 2023). If a probe decodes these phases from residual-stream activations, it is tempting to say that the model represents an internal reasoning state: *I am thinking now; I am acting now; I am observing now*. Yet phase also predicts the kind of text at the current position. Thoughts contain explanations and calculations, actions contain calls and commands, and observations contain structured results. Markers, local position, content type, and phase role are therefore entangled by construction.

We study this ambiguity in matched Gemma-2 base and instruction-tuned (IT) pairs at 2B and 9B scale (Gemma Team, 2024). The project began with a result that looked like evidence for an abstract phase variable. In natural synthetic traces, phase accuracy peaks broadly at mid-depth, the base and IT curves nearly overlap, and removing marker literals and newlines barely changes the peak. Cross-format marker transfer is also strong. Those controls establish that the probe is not merely classifying the current marker token. They do not counterbalance phase-correlated content or the broader positional and template structure of a trace.

Our confirmatory test constructs a decorrelation ladder. A procedural dataset crosses each inner-loop phase with mathematics, code/structured data, and prose; varies phase order and repetition; and must pass frozen lexical, surface, content-balance, and position-only gates before model activations are examined. We normalize accuracy by a matched natural-trace denominator, defining retained signal R, then remove markers and newlines in successive rungs. Thresholds for an illusion band, an abstract band, and an IT/base invariance margin were fixed before the confirmatory runs.

The result changes the scientific story. At 2B, the marked decorrelated set preserves a large signal and a large IT advantage. Once marker literals

are removed from the same content-and-position-controlled traces, both point estimates fall into the preregistered illusion band and the IT–base gap shrinks to 0.018 with its interval crossing zero. At 9B, both stripped signed-gap intervals also cross zero, but retention sits at the illusion-band boundary. The defensible cross-scale synthesis is not that every cutoff replicates. It is that the apparent phase variable is strongly cue-dependent, and that IT/base invariance after cue removal is stable across the two tested scales.

Our contributions are threefold: (1) a preregistered content/position/marker ladder with matched denominators and explicit controls; (2) a 2B-to-9B replication that separates robust IT/base invariance from scale-dependent threshold details; and (3) SAE feature analysis plus bounded patching that narrows what the natural-trace probe can mean. We do not claim that phase is nowhere represented. We show that near-perfect natural decoding is insufficient evidence for a clean abstract phase variable.

2 Related Work

Probing and controls. Probe accuracy can conflate information in a representation with structure learned by the probe or correlations in the evaluation data. Control tasks and selectivity place accuracy in context (Hewitt and Liang, 2019); controlled datasets show that high probe accuracy need not entail relevance to the model's task (Ravichander et al., 2021); and amnesic probing distinguishes encoded from behaviorally used information (Elazar et al., 2021). Belinkov (2022) surveys these limitations. Our intervention is at the data-design level: we hold the probe family fixed and remove plausible label-correlated cues.

Reasoning traces and faithfulness. Chain-of-thought prompting elicits intermediate reasoning text (Wei et al., 2022), while ReAct interleaves reasoning with actions and observations (Yao et al., 2023). Verbalized reasoning is not necessarily a complete account of answer-producing computation (Lanham et al., 2023; Turpin et al., 2023; Arcuschin et al., 2025), although definitions and measurements of faithfulness remain contested (Zaman and Srivastava, 2025). We test a prior interpretive step: whether decoding the *format phase* identifies an abstract internal reasoning state.

Sparse features and interventions. SAEs provide sparse dictionaries for decomposing activations (Bricken et al., 2023); Gemma Scope releases layerwise JumpReLU SAEs for Gemma 2 (Lieberum et al., 2024). Activation patching can test whether a selected direction is causally efficacious, but its conclusion depends on the intervention and metric (Heimersheim and Nanda, 2024) and should be distinguished from a full causal abstraction claim (Geiger et al., 2023).

3 Experimental Design

3.1 Models, representations, and labels

We compare `gemma-2-2b` with `gemma-2-2b-it`, and `gemma-2-9b` with `gemma-2-9b-it`. Within each scale, base and IT use the same tokenizer. The 2B pair has zero mismatches or unknown tokens over 525 stored trace variants; the combined token-id hash matches exactly. The 2B and 9B tokenizers also produce identical ids on all decorrelation rungs.

We cache residual streams after each transformer block (`resid_post`). Natural analyses label Question, Thought, Action, Observation, Answer, and Other. We report ordinary token accuracy for exploratory E1/E2 and balanced accuracy for the three-way confirmatory analyses; $\mathrm{acc}_{\mathrm{dec}}$ and $\mathrm{acc}_{\mathrm{nat,matched}}$ below denote balanced accuracy. Confirmatory analysis is balanced over Thought/Action/Observation (T/A/O), excludes tokens inside marker literals, and includes only tokens inside labeled inner-loop content segments. Prefix exclusion prevents direct classification of the current marker token, but marked traces still make marker context available.

All traces are synthetic or procedurally generated. Observation text is fabricated plausible output rather than a record of real tool execution. The data support claims about phase-label decodability under controlled surface forms, not factual correctness or real-world tool use.

3.2 Natural-trace setup: E1 and E2

E1 fits a multinomial linear probe at every 2B layer for base and IT under three forms: marked, marker-stripped, and marker-plus-newline-stripped. Five seeds use an 80/20 stratified split and per-fold standardization. This exploratory analysis establishes the phenomenon motivating stronger controls.

E2 changes inner-loop marker vocabulary while holding content fixed, in both train/evaluation directions. Transfer gap is in-distribution minus cross-format accuracy, so a smaller gap is better. A leak-

age mask removes target positions before the first changed marker and any target residual exactly equal to its content twin at the same position. Only the marked condition is genuine marker-format transfer; after marker removal the matched texts are identical.

3.3 Preregistered decorrelation ladder

The E5b dataset contains 45 traces. It crosses each of T/A/O with mathematics, code or structured data, and prose, while varying inner-loop order, count, and repetition. Content balance and lexical and position-only classifiers are evaluated before activations are cached.

All frozen gates pass. Maximum content total variation is 0.003. The lexical baseline has mean/max balanced accuracy 0.327/0.397. The position battery has maximum mean/max-seed accuracy 0.365/0.391, and a stronger character-ngram plus surface-statistics control reaches 0.322 token-weighted balanced accuracy. Each is below the registered 0.45 gate. These controls do not prove perfect independence, but they rule out the measured simple alternatives.

The neural probe is multinomial logistic regression optimized with SciPy L-BFGS-B ($C = 1$, maximum 1,000 iterations), with a StandardScaler fit inside each fold, an 80/20 stratified token split, and seeds 42–46. This preregistered split estimates within-dataset decodability on the globally content- and position-gated sample but permits within-trace sharing, so trace-specific cues may raise R; it is not an unseen-trace generalization test. Appendix C therefore adds an exploratory whole-trace split. Because some grouped folds exceed the registered content-balance gate, that analysis bounds generalization rather than replacing the controlled headline estimand. The 2B headline layers are 12 and 13. The 9B replication fixes layers 20 and 21 by matching relative depth before inspecting 9B results.

We define retained signal as

$$R = \frac{\mathrm{acc}_{\mathrm{dec}} - 1/3}{\mathrm{acc}_{\mathrm{nat,matched}} - 1/3}, \qquad (1)$$

where the natural denominator contains 4,547 tokens per phase. Thus $R = 0$ is chance and $R = 1$ is full retention relative to matched natural traces. The preregistration labels $R \leq 0.33$ the positional/content *illusion* band, $0.33 < R < 0.66$ partial or ambiguous, and $R \geq 0.66$ the abstract range, subject to neural accuracy exceeding the strongest control by more than 0.10. IT/base invariance is $|R_{\mathrm{IT}} - R_{\mathrm{base}}| \leq 0.10$. We emphasize continuous estimates near a boundary. These thresholds, the headline-layer rule, and the control gates were committed to version control as a frozen preregistration before any confirmatory activations were probed, and the 9B replication reused this preregistration without modification.

The ladder evaluates the same traces with (i) marker literals present, (ii) markers removed, and (iii) markers and inter-segment newlines removed. Because prefix tokens are excluded, marked-versus-unmarked differences measure marker-literal/context availability, not direct classification of the literal tokens.

3.4 Cross-scale replication and uncertainty

The 9B run reuses the same trace text, ablations, thresholds, probe configuration, and retention rule. No data are regenerated and no headline layer is selected post hoc. Token counts are 25,399, 24,769, and 24,499 across the ladder, with exact tokenizer parity.

Two frozen uncertainty summaries are available. The 9B preregistration assigns 10,000 seed-only replicates to per-model R and 5,000 paired trace-and-seed replicates to signed gaps. For common trace-level bars, Figure 2 additionally shows per-model R distributions already computed by the paired routine—a disclosed reporting harmonization, not new computation, that changes no point estimate or verdict. Paired replicates jointly resample the same 45 traces and five seeds across models, recomputing R and gaps from trace-weighted cached held-out predictions and seed-resampled natural denominators without refitting probes. Both routines ran unchanged at 9B.

3.5 Feature-level corroboration: E3 and E4

E3 encodes 2B activations with Gemma Scope residual SAEs at six layers. No Gemma-2-2B-IT residual SAE is released, so the cross-model analysis applies the base SAE to both physical models. Activations are standardized within each model's T/A/O pool before taking an IT-minus-base difference of phase differentials. A random-label null repeats the comparison after permuting phase labels.

We classify released explanations of the base-ranked top ten layer-12 features per phase as content-domain, process-role, ambiguous, or unavailable. E4 patches selected SAE directions across phases at layers 12, 16, and 20. The out-

come is the change in source-minus-target probability mass over phase-preferred non-marker content tokens; matched random live features form the primary null. This bounded intervention is not a test of a complete reasoning circuit.

4 Results

4.1 Natural traces create the appearance of abstraction

Figure 1 shows E1. With markers, base peaks at layer 13 with accuracy 0.988 and IT at layer 12 with accuracy 0.991. Both show a broad middle-layer peak. More strikingly, after markers and newlines are removed, layer-12 accuracy is 0.979 for both and the peak remains 0.979–0.981. The mean across-layer drop is only 0.013 for base and 0.016 for IT.

E2 reinforces the initial appearance. After leakage masking, the forward L12 IT-minus-base transfer-gap difference is $+0.0001 \pm 0.0017$; in reverse it is -0.0142 ± 0.0023, a small IT advantage because lower transfer gap is better. These results rule out a pure current-marker-token explanation. They do not decorrelate phase from segment content, local context, or position. E1 and E2 are therefore the setup for the interpretability illusion, not evidence that resolves it.

4.2 At 2B, markers carry the decorrelated gap

The preregistered ladder gives a different result (Figure 2, Table 1). On marked decorrelated traces, base retains $R = 0.630$ $[0.601, 0.659]$, in the partial band, while IT retains $R = 0.778$ $[0.758, 0.797]$, in the abstract range. The signed gap is 0.148 $[0.123, 0.173]$, excluding zero and the 0.10 margin. It is positive in 25 of 26 layers and robust to dropping any probe seed.

Removing markers changes the verdict. Base falls to $R = 0.276$ $[0.237, 0.314]$ and IT to $R = 0.295$ $[0.259, 0.331]$; balanced accuracies are 0.515 and 0.528. Both point estimates lie in the registered illusion band, although the IT interval narrowly crosses the 0.33 boundary, and the gap is 0.018 $[-0.006, 0.043]$. Removing newlines as well gives $R_{\text{base}} = 0.281$ $[0.244, 0.318]$, $R_{\text{IT}} = 0.291$ $[0.258, 0.325]$, and gap $+0.010$ $[-0.016, +0.038]$.

Marker availability adds 0.232 balanced-accuracy points for base and 0.318 for IT on the decorrelated set, an extra IT gain of 0.087. Since current marker tokens are excluded, this is marker-context reliance. The registered 2B conclusion is that marker-robust abstract-phase survival is unsupported. The robust IT difference is greater exploitation of marked phase context, not a marker-free abstract phase representation.

4.3 9B preserves invariance, not every cutoff

At 9B, the marked rung gives $R_{\text{base}} = 0.663$ and $R_{\text{IT}} = 0.734$, with gap $+0.070$ $[+0.047, +0.096]$. Neural accuracy exceeds the strongest control by +0.400 for base and +0.446 for IT. Unlike 2B, the marked gap remains below the frozen 0.10 margin. The large 2B IT marker-reliance difference therefore does not replicate.

After marker removal, the comparison is tighter: $R_{\text{base}} = 0.325$, $R_{\text{IT}} = 0.328$, gap $+0.003$ $[-0.023, +0.028]$. Removing newlines yields $R_{\text{base}} = 0.337$, $R_{\text{IT}} = 0.331$, gap -0.006 $[-0.030, +0.017]$. Both signed intervals include zero and are far inside the invariance margin.

The threshold detail matters. The no-marker points are in the illusion band, but the no-newline base point is 0.337, just above 0.33, and both per-model intervals straddle that boundary. The mechanical statement "both models are in the illusion band" does not fully replicate. We do not move the threshold. The continuous pattern is clear: marker removal roughly halves retention, and the stripped IT/base gap is negligible at both scales. Cue dependence recurs; IT/base invariance generalizes; the exact illusion-band verdict does not. A held-out-trace robustness check at 2B (Appendix C) reinforces the collapse: under whole-trace cross-validation the marker-free signal does not generalize (stripped-regime retention falls to near chance, $R \approx 0$), while the IT–base point gap stays within the invariance margin.

4.4 Feature evidence supports content entanglement

Among the base-ranked top ten layer-12 SAE features for each phase, 28 of 30 slots (24 unique ids) are content-domain and two are process-role (Figure 3a). Action is 10 of 10 content-domain; Thought and Observation include mathematics, programming, structured data, identifiers, logs, and statistical formats. These labels come from released automated feature explanations rather than manual inspection, so we read the 28-of-30 count as indicative of content entanglement rather than a definitive taxonomy.

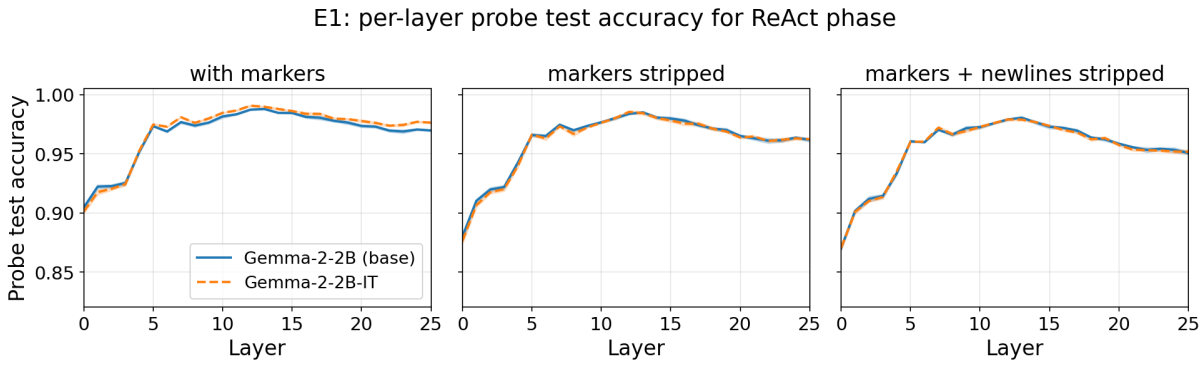


Figure 1: Natural-trace phase-probe accuracy (base solid, IT dashed). Marker and newline removal preserves near-ceiling decoding because natural templates retain phase-correlated content and sequence structure.

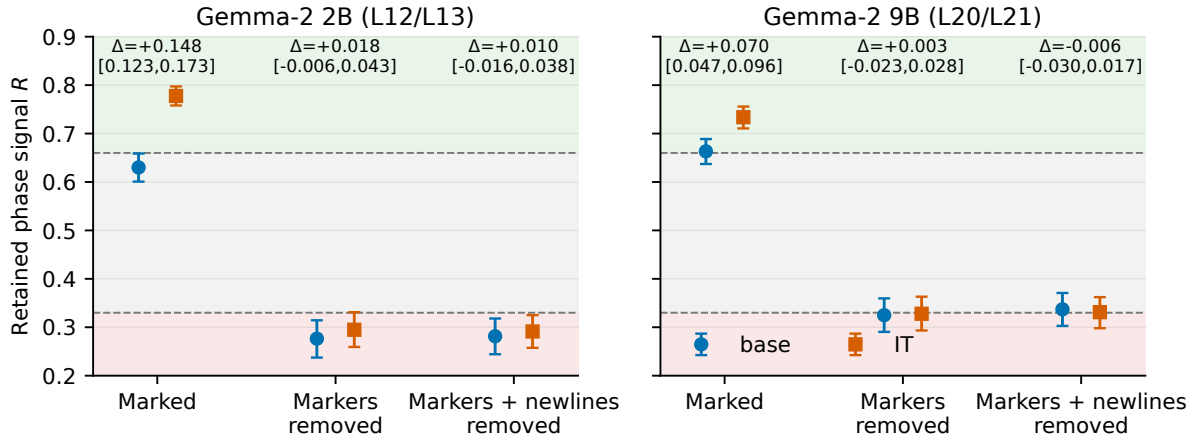


Figure 2: Cross-scale retained-signal ladder at fixed relative-depth layers. Error bars are 95% paired trace-and-seed bootstrap intervals. Background bands are the frozen illusion ($R \leq 0.33$), partial, and abstract ($R \geq 0.66$) regions. Δ is signed IT-minus-base retention from the same replicates.

No shared-basis layer/phase cell crosses the preregistered reorganization bar: median absolute IT-minus-base differentials range from 0.005 to 0.013 standard deviations against a 0.10 bar, and the largest elevated-feature count is 14 where roughly 130–150 are required. Under permuted phase labels, the median floor is 0.003–0.008 and elevated-feature count is zero everywhere.

E4 shows that selected directions are not entirely inert. Pooled real cross-phase patches exceed random-live-feature patches by $+0.000353$ $[0.000127, 0.000613]$ in source-minus-target content-token mass. The effect is small and unstable across layers: -0.000092 at L12, $+0.000339$ at L16, and $+0.000812$ at L20. Dropping repeat seed 60429 yields $+0.000229$ $[-0.000001, +0.000464]$. IT is not larger than base: IT-minus-base is -0.000079 $[-0.000444, +0.000268]$.

E3 therefore shows that inspected marked-trace directions mainly track what kind of content appears in a phase. E4 shows that those directions can slightly move next-token distributions under the tested intervention, but does not establish an abstract phase circuit. An earlier current-token/bucket balance left L12 accuracy high (0.9897 base, 0.9899 IT), illustrating why full segment-level decorrelation was needed.

5 Discussion

Why natural marker stripping was insufficient. The natural and decorrelated ablations appear contradictory only if deletion is treated as decorrelation. Natural traces still place characteristic content in characteristic phases, preserve a regular segment cycle, and reuse phase-specific local templates. A marker-free action still looks like a call; a marker-free observation still looks like a result. E5b crosses those content types with all phases and varies their positions. Only then does marker

Scale	Rung	R_{base}	R_{IT}	Signed IT–base gap [95% interval]
2B	marked	0.630	0.778	$+0.148\ [+0.123, +0.173]$
2B	no markers	0.276	0.295	$+0.018\ [-0.006, +0.043]$
2B	no markers/newlines	0.281	0.291	$+0.010\ [-0.016, +0.038]$
9B	marked	0.663	0.734	$+0.070\ [+0.047, +0.096]$
9B	no markers	0.325	0.328	$+0.003\ [-0.023, +0.028]$
9B	no markers/newlines	0.337	0.331	$-0.006\ [-0.030, +0.017]$

Table 1: Headline matched-denominator retention. Figure 2 plots per-model intervals; signed-gap intervals use the same paired trace-and-seed bootstrap.

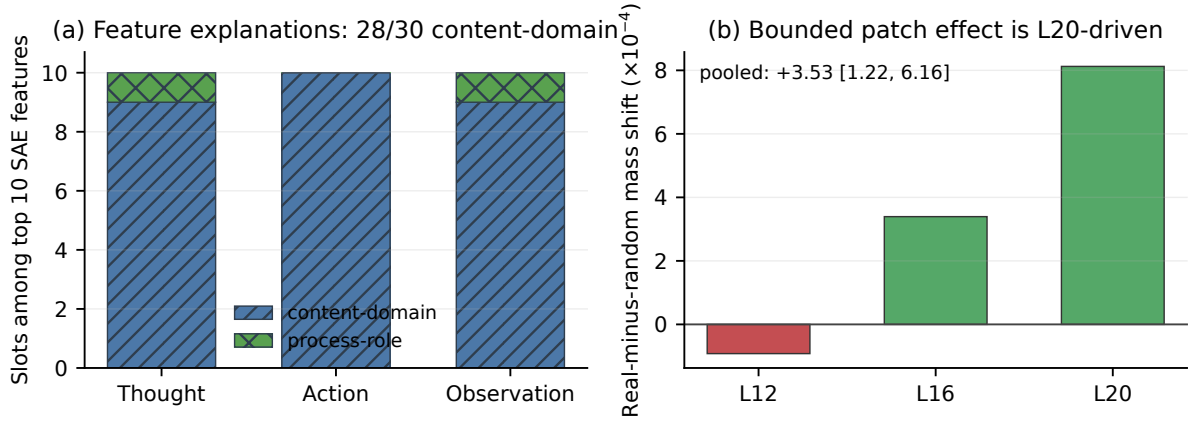


Figure 3: Feature-level corroboration at 2B. (a) Released explanations for top phase-selective SAE features are overwhelmingly content-domain. (b) The bounded real-minus-random patch effect is pooled and L20-driven. Layer bars are descriptive; the annotation gives the pooled bootstrap interval in units of 10^{-4}.

removal test the residual variable after dominant alternatives are suppressed. Surface ablations can make a signal look "semantic" while leaving the semantics of the nuisance variable intact.

What instruction tuning changes. The marked 2B result could be read as IT creating a more abstract phase representation: IT retains more signal and crosses the abstract threshold. The stripped rungs do not support that reading. The gap shrinks and its interval crosses zero when marker context is removed, and stripped-regime invariance holds at 9B. The narrow positive result is about cue use: 2B IT extracts more phase information from marker availability than 2B base. That difference is scale-dependent. We do not show that instruction tuning never reorganizes reasoning. We show no robust base/IT separation for the tested marker-free, content-and-position-controlled phase signal.

Recommended standard. Before naming a decoded direction as an abstract process variable, probe studies should (i) counterbalance label-correlated content, (ii) test position and surface classifiers before neural analysis, (iii) use cue ladders and matched denominators, and (iv) distinguish decodability from use with interventions whose null, layer sensitivity, and effect size are reported. Each step targets a distinct illusion mode: content counterbalancing removes the dominant nuisance variable, surface and position baselines bound trivial explanations, matched denominators turn raw accuracy into an interpretable retention scale, and reported intervention statistics keep causal claims proportionate to their evidence. In our study only the full ladder—not any single ablation—separated a cue-dependent signal from a candidate abstract variable, and only the cross-scale run distinguished a robust invariance from a scale-specific threshold artifact.

6 Conclusion

ReAct phase is almost perfectly decodable from natural Gemma-2 residual streams and remains so under simple marker and newline ablations. A pre-registered decorrelation ladder shows why that is not enough to infer an abstract reasoning-state variable. At 2B, marker removal collapses retention and reduces the IT advantage to a gap whose interval crosses zero; at 9B, cue dependence and stripped-regime IT/base invariance recur, while the strict illusion cutoff and large marked gap do not.

SAE features and bounded patching corroborate content entanglement at 2B. Phase probes decode a mixture of process label and correlated cues; claims of abstraction should be earned by decorrelation, controls, and interventions rather than near-ceiling natural accuracy.

Limitations

The traces are synthetic and procedural. Their observations are fabricated plausible text, and we do not measure factual correctness, real tool use, or answer faithfulness.

The confirmatory set contains 45 traces from one generator and one three-way content taxonomy. Its controls pass the registered gates, but no finite battery proves independence from all lexical or structural cues. Token-level stratified splits can share templates across train and test. An exploratory whole-trace held-out analysis (Appendix C) shows the marker-free signal does not generalize to held-out traces (retention near chance) and the IT/base point gap stays within the invariance margin; because those folds are not content-balanced it bounds generalization rather than isolating a trace-template mechanism, and it still resamples the same 45 traces rather than testing a held-out task.

The cross-scale comparison covers two sizes in one model family. The 9B result shows why this matters: invariance generalizes while the strict cutoff does not. The R bands are preregistered operational thresholds, not natural kinds.

At 2B, no IT-specific Gemma Scope SAE exists, so E3/E4 use a shared base-SAE basis. Standardization and a random-label null limit but do not erase basis mismatch. E4 is small, L20-driven, sensitive to one repeat seed, and does not show that patched directions are used for reasoning. Finally, we probe externally supplied trace text, not internal states during online traces generated by the same model.

Ethics Statement

This work analyzes synthetic and procedurally generated ReAct traces. It involves no human subjects, no private or personally identifiable data, and no model training beyond lightweight linear probes on cached activations. Because all observations are fabricated plausible text, the paper makes no claims about real tool use or factual correctness. The intended impact is methodological: by showing that near-ceiling probe accuracy can reflect content and positional cues rather than an abstract reasoning state, we aim to reduce over-interpretation of probing results in interpretability research. We identify no direct risk of harmful application.

References

Iván Arcuschin, Jett Janiak, Robert Krzyzanowski, Senthooran Rajamanoharan, Neel Nanda, and Arthur Conmy. 2025. Chain-of-thought reasoning in the wild is not always faithful. *arXiv preprint arXiv:2503.08679*.

Yonatan Belinkov. 2022. Probing classifiers: Promises, shortcomings, and advances. *Computational Linguistics*, 48(1):207–219.

Trenton Bricken, Adly Templeton, Joshua Batson, Brian Chen, Adam Jermyn, Tom Conerly, Nick Turner, Cem Anil, Carson Denison, Amanda Askell, Robert Lasenby, Yifan Wu, Shauna Kravec, Nicholas Schiefer, Tim Maxwell, Nicholas Joseph, Zac Hatfield-Dodds, Alex Tamkin, Karina Nguyen, and 6 others. 2023. Towards monosemanticity: Decomposing language models with dictionary learning. *Transformer Circuits Thread*.

Yanai Elazar, Shauli Ravfogel, Alon Jacovi, and Yoav Goldberg. 2021. Amnesic probing: Behavioral explanation with amnesic counterfactuals. *Transactions of the Association for Computational Linguistics*, 9:160–175.

Atticus Geiger, Duligur Ibeling, Amir Zur, Maheep Chaudhary, Sonakshi Chauhan, Jing Huang, Aryaman Arora, Zhengxuan Wu, Noah Goodman, Christopher Potts, and Thomas Icard. 2023. Causal abstraction: A theoretical foundation for mechanistic interpretability. *arXiv preprint arXiv:2301.04709*.

Gemma Team. 2024. Gemma 2: Improving open language models at a practical size. *arXiv preprint arXiv:2408.00118*.

Stefan Heimersheim and Neel Nanda. 2024. How to use and interpret activation patching. *arXiv preprint arXiv:2404.15255*.

John Hewitt and Percy Liang. 2019. Designing and interpreting probes with control tasks. In *Proceedings of the 2019 Conference on Empirical Methods in Natural Language Processing and the 9th International Joint Conference on Natural Language Processing (EMNLP-IJCNLP)*, pages 2733–2743, Hong Kong, China. Association for Computational Linguistics.

Tamera Lanham, Anna Chen, Ansh Radhakrishnan, Benoit Steiner, Carson Denison, Danny Hernandez, Dustin Li, Esin Durmus, Evan Hubinger, Jackson Kernion, Kamilė Lukošiūtė, Karina Nguyen, Newton Cheng, Nicholas Joseph, Nicholas Schiefer, Oliver Rausch, Robin Larson, Sam McCandlish, Sandipan

Kundu, and 11 others. 2023. Measuring faithfulness in chain-of-thought reasoning. *arXiv preprint arXiv:2307.13702*.

Tom Lieberum, Senthooran Rajamanoharan, Arthur Conmy, Lewis Smith, Nicolas Sonnerat, Vikrant Varma, János Kramar, Anca Dragan, Rohin Shah, and Neel Nanda. 2024. Gemma scope: Open sparse autoencoders everywhere all at once on gemma 2. *arXiv preprint arXiv:2408.05147*.

Abhilasha Ravichander, Yonatan Belinkov, and Eduard Hovy. 2021. Probing the probing paradigm: Does probing accuracy entail task relevance? In *Proceedings of the 16th Conference of the European Chapter of the Association for Computational Linguistics: Main Volume*, pages 3363–3377, Online. Association for Computational Linguistics.

Miles Turpin, Julian Michael, Ethan Perez, and Samuel R. Bowman. 2023. Language models don't always say what they think: Unfaithful explanations in chain-of-thought prompting. *arXiv preprint arXiv:2305.04388*. NeurIPS 2023.

Jason Wei, Xuezhi Wang, Dale Schuurmans, Maarten Bosma, Brian Ichter, Fei Xia, Ed H. Chi, Quoc V. Le, and Denny Zhou. 2022. Chain-of-thought prompting elicits reasoning in large language models. *arXiv preprint arXiv:2201.11903*.

Shunyu Yao, Jeffrey Zhao, Dian Yu, Nan Du, Izhak Shafran, Karthik Narasimhan, and Yuan Cao. 2023. React: Synergizing reasoning and acting in language models. *arXiv preprint arXiv:2210.03629*. ICLR 2023 camera-ready version.

Kerem Zaman and Shashank Srivastava. 2025. Is chain-of-thought really not explainability? chain-of-thought can be faithful without hint verbalization. *arXiv preprint arXiv:2512.23032*.

A Additional Natural-Transfer Details

The E2 leakage mask removes positions before the first swapped marker and target residuals exactly equal to the matched source residual at the same position. It masks 13.8–14.3% of L12 target tokens. Masked forward gaps are 0.0394 for both models; reverse gaps are 0.0286 for base and 0.0144 for IT. Marker-stripped content twins are identical texts, so only marked E2 is interpretable as transfer.

B Reproducibility Summary

Probe fits use frozen model ids, hook names, seeds, splits, scalers, and regularization. E1 contains 780 model/variant/layer/seed fits. A cache-schema rebuild adding the TransformerLens version reproduced all 780 accuracies with maximum absolute difference 0.0. The 9B run records tokenizer parity, cache hook assertions, versions, data hashes, and an integrity guard over frozen 2B inputs. A claim ledger maps each reported number to a tracked JSON or generated findings artifact. All code, data, preregistration files, and the claim ledger will be released publicly upon acceptance.

	R_{base}		R_{IT}	
Rung (2B)	token	trace	token	trace
marked	0.630	0.426	0.778	0.631
no markers	0.276	−0.02	0.295	0.01
no mk./newlines	0.281	−0.01	0.291	0.01

Table 2: Retained signal R at L12/L13 under the preregistered stratified token split vs. an exploratory whole-trace held-out split (2B). Stripped-regime signed IT–base gaps stay within the 0.10 invariance margin (+0.03, +0.02); marked retention drops but the IT edge persists. Grouped folds are not content-balanced (max phase-to-pooled TV $0.158 > 0.10$ gate; see text).

C Held-Out-Trace Robustness

The confirmatory probe uses a stratified token split, so tokens from a single trace can fall in both train and test. As an exploratory check (not part of the preregistration), we rerun the identical probe—same balanced subsampling, standardization, L-BFGS multinomial fit, and seeds—but replace the token split with a whole-trace held-out split, so every token of a trace lies on one side of the fold. The token arm reproduces the frozen headline decode accuracy at layers 12–13 for all three rungs, so the two arms are directly comparable. Table 2 reports retained signal R under both splits at 2B. Held-out-trace evaluation lowers marked retention (R_{base} $0.630 \to 0.43$, R_{IT} $0.778 \to 0.63$) and, in both stripped regimes, drives it to near chance ($R \approx 0$, below the illusion band), while the signed IT–base gap stays within the 0.10 invariance margin (+0.03, +0.02). These whole-trace folds are not content-balanced—some folds exceed the registered content-balance gate (maximum phase-to-pooled content total variation 0.158 vs. the 0.10 gate used for the main analysis)—so this check bounds held-out-trace generalization rather than isolating a purely trace-template mechanism from residual content imbalance. Read that way, it reinforces the illusion collapse and the stripped-regime IT/base invariance under a stricter split.